



AI-Assisted EEG Screening for Rapid & Informed Critical Care

TCS-Net: Methodology, Results & Comparative Analysis

A Multimodal Deep Learning Approach to Harmful Brain Activity Classification

Master's Project: Final Report

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Abstract

Continuous EEG monitoring is standard in neurological ICUs, but its clinical value is bottlenecked by the need for trained neurophysiologists to interpret long recordings in real time. This project investigates whether modern deep learning can act as a screening assistant, flagging harmful brain activity patterns (seizures, periodic discharges, rhythmic activity) with calibrated probabilities that respect the genuine disagreement among expert annotators.

We developed TCS-Net (Temporal Cross-Spectrogram Network), a 29.6M-parameter multimodal architecture that fuses raw EEG encoding with dual-source spectrogram processing through a novel Cross-Modal Token Attention (CMTA) mechanism. To evaluate it fairly we re-trained two contrasting baselines: VIPEEGNet (npj Digital Medicine, 2025) and a Deotte-style EfficientNetB0 on a single shared 5-fold patient-grouped split, so all three models validate on the same patients in each fold.

On the high-quality subset (5,939 EEG-IDs with ≥ 10 expert votes), TCS-Net reaches an out-of-fold KL divergence of 0.376, between VIPEEGNet (0.220) and EfficientNetB0 (0.673). However, our central finding is that KLD alone is misleading: extending the evaluation across AUROC/AUPRC, operating-point cost, calibration, stability, hard-miss curves, soft-label fidelity, confidence stratification, and per-class IIC structure reveals systematically different strengths. VIPEEGNet wins essentially every accuracy axis (AUROC 0.946, AUPRC 0.667) and is well-calibrated raw. EfficientNetB0 is severely miscalibrated (ECE 0.108) and must ship with per-fold Platt scaling (ECE $\rightarrow$ 0.025). TCS-Net wins only raw threshold stability (1.6 $\times$ spread vs. 13 $\times$ for VIPEEGNet) and is otherwise dominated, twice the parameters and 3 $\times$ the training time of VIPEEGNet, lower per-class F1 across the board. The refined recommendation is VIPEEGNet primary, EfficientNetB0 + Platt as backup, and TCS-Net retained as a research artefact. The multi-metric evaluation framework that produced these distinctions is the main reusable contribution.

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1. Background and Motivation

1.1 Clinical Problem

Patients in neurological intensive care frequently undergo continuous EEG monitoring for hours to days at a time. The recordings can capture life-threatening events such as non-convulsive status epilepticus, periodic discharges that signal rising intracranial pressure, or rhythmic delta activity that correlates with delayed cerebral ischemia. Acting on these patterns quickly improves outcomes; missing them can cause irreversible neurological injury.

The bottleneck is human attention. A single 24-hour EEG recording produces roughly two million sample points across ~19 electrodes. Reviewing it requires a board-certified neurophysiologist working through the trace at fast playback while looking for patterns that may occupy only a few seconds in each hour. Larger hospitals have an on-call EEG service that reviews recordings every few hours; smaller community ICUs either send recordings out for batched overnight review or have no dedicated EEG coverage at all. Even at top centers, time-to-detection for non-convulsive seizures is measured in hours, not minutes.

An automated system that screens EEG continuously and produces calibrated probabilities for the six clinically relevant patterns: Seizure, LPD (Lateralized Periodic Discharges), GPD (Generalized Periodic Discharges), LRDA (Lateralized Rhythmic Delta Activity), GRDA (Generalized Rhythmic Delta Activity), and Other could reduce time-to-detection from hours to seconds, deliver consistent results across reviewers and shifts, and free human experts to focus on the cases the model is uncertain about. The clinical value is not in replacing the neurophysiologist; it is in choosing what they look at first.

1.2 What Makes This Hard

Three properties of the EEG classification problem make it harder than it looks at first.

- **Soft, disagreement-laden labels.** Every recording in our dataset is annotated by approximately 12 neurologists who vote independently for one of the six classes. The ground-truth label is the empirical vote distribution, not a single class. A 7 vs 3 split between Seizure and LPD is qualitatively different from a 10-0 Seizure label, and a model that predicts a hard one-hot answer has lost information the experts gave it. The competition metric (KL divergence between predicted and target distributions) penalizes both overconfidence on disagreement cases and underconfidence on consensus cases.
- **Bimodal label quality.** The number of votes per recording is bimodal: most segments either have 1-7 votes (noisy, single annotator-cluster) or ≥ 10 votes (reliable expert consensus), with a near-empty gap at 8-9. Training only on high-quality segments leaves about 36,000 examples; training on everything pulls the model toward noisy labels. A staged training strategy that warms up on all data and fine-tunes on the high-quality subset is therefore the dominant pattern in the literature.
- **Patient-level leakage.** Most patients contribute multiple recordings. A naive random split puts the same patient in train and validation, and the model gets credit for memorizing patient-specific

features (electrode placement, baseline rhythm, age-related slowing). Patient-grouped cross-validation is mandatory; without it, validation KLD is silently optimistic by 0.05-0.10.

1.3 Related Work

This project draws on three converging lines of work: the HMS Kaggle competition writeups (which represent the engineering state of the art), recent peer-reviewed EEG deep learning models (which represent the architectural state of the art), and the broader machine-learning literature on calibration and multimodal fusion.

1.3.1 The HMS Competition (Harvard Medical School, 2024)

The HMS - Harmful Brain Activity Classification competition (Jing et al., 2024) released the dataset used [here](#) and produced over 2,500 public submissions. We analyzed the publicly disclosed methodologies for the top 10 finishers. Several patterns held across nearly every gold-medal solution: dual spectrogram input combining the Kaggle-provided 10-minute spectrograms with EEG-derived 50-second spectrograms; EfficientNet-family backbones (B0, B2, V2-S); two-stage training that first fits all data with vote-weighted loss and then fine-tunes only on the high-quality subset; standard EEG augmentations (brain-flip, time-reverse, channel dropout); and exponential moving averages of model weights. The 1st place solution (Team Sony, 0.216 KLD) ensembled 30 different backbones at inference; the 2nd place (COOLZ, 'Holiday', 0.219) used a 3D-CNN over multi-channel spectrograms; the 8th place (Chris Deotte, 0.24) was a single mega-model that became the most-replicated baseline in the public Kaggle ecosystem. None of the top solutions used cross-modal attention: every one of them processed EEG and spectrograms independently and concatenated the resulting feature vectors before the classifier head.

1.3.2 VIPEEGNet (npj Digital Medicine, 2025)

VIPEEGNet (Sunyuri et al., 2025) is the strongest single peer-reviewed model on the HMS task. Its central idea is to convert the raw 16-channel bipolar EEG into a pseudo-image using three stacked Conv1D layers with a SumToOne weight constraint, the kernel weights along the input-channel dimension are softmax-normalized so that they are strictly positive and sum to one. This guarantees that each output channel is a proper convex combination of the input bipolar channels, preventing the model from learning channel weightings that exploit training-set artifacts. The pseudo-image is then fed to a standard EfficientNetV2-B3 backbone. VIPEEGNet reports 0.225 OOF KLD on a 10-fold patient-grouped split, with only 13M parameters.

1.3.3 The Deotte Spectrogram Recipe

Chris Deotte (Kaggle, 2024) reverse-engineered the exact librosa parameters that produce the Kaggle-provided spectrograms (`n_fft=1024`, `hop_length=128`, `n_mels=128`, frequency range 0–20 Hz, log-power compression, per-channel normalization). Using generic scipy STFT with different parameters produces spectrograms that look visually identical to a human but differ enough in texture statistics to confuse

ImageNet-pretrained backbones. In our preliminary runs, swapping correct Deotte parameters for naive scipy parameters increased OOF KLD from 0.34 to 0.82, a single preprocessing decision that mattered more than most architectural choices.

1.3.4 Calibration and Cross-Modal Fusion

Two pieces of broader ML literature directly inform the project. Guo et al. (ICML 2017) showed that modern neural networks trained with cross-entropy are systematically overconfident, and that learning a single scalar temperature parameter on a held-out set substantially improves calibration measured by ECE (Expected Calibration Error). For tasks where the target is a probability distribution rather than a class exactly our situation, calibration error directly drives KLD, but no published HMS solution applied temperature scaling. SpecAugment (Park et al., Interspeech 2019) introduced frequency and time masking as a regularizer for speech recognition; the 2nd, 3rd, and 4th place HMS solutions adapted it to EEG spectrograms, and we follow them.

1.4 What This Project Adds

Building on this body of work, the project makes four contributions:

- **TCS-Net architecture.** A multimodal model that, unlike every published HMS solution, uses bidirectional Cross-Modal Token Attention between the EEG temporal stream and the spectrogram patch stream, plus a Frequency Band Gate that uses EEG context to dynamically reweight the four canonical EEG frequency bands (delta, theta, alpha, beta) of the spectrogram features.
- **Six targeted training-protocol fixes** (SpecAugment, random temporal shift, extended schedule, SumToOne stem, per-fold temperature scaling, attempted checkpoint averaging) that together improved OOF KLD from 0.341 to 0.329 with no architectural change. Temperature scaling alone never previously applied to HMS accounted for ~ 0.012 of that improvement at zero retraining cost.
- **A fair-comparison evaluation framework** with shared 5-fold patient splits across all three models (TCS-Net, VPEEGNet, EfficientNetB0). Every model validates on the same patients in each fold; OOF predictions are saved in a unified format and analyzed downstream.
- **An evaluation suite that goes beyond KLD:** AUROC and AUPRC with bootstrap confidence intervals, operating-point cost at clinically relevant recall thresholds, alert-burden curves, expected calibration error, fold-wise threshold stability, hard-miss enumeration, and ambiguity-conditioned error analysis. This is the part of the project that turned out to matter most: it shows that the right model depends on the deployment scenario, not on a single leaderboard number.

2. The TCS-Net Architecture

2.1 Design Hypothesis

Every top-10 HMS solution we surveyed encoded EEG and spectrograms with two parallel branches and concatenated their pooled features just before the classifier head. The implicit assumption is that EEG features and spectrogram features are independent enough that simply joining them is sufficient. We hypothesized that this is wrong for clinically important pattern, it often requires correlating where in the EEG a pattern lives with what frequency content the spectrogram shows there. A late-fusion concatenation cannot do that correlation; the two branches never talk to each other before the head. TCS-Net's response is to introduce a small number of cross-attention layers between the two encoders. The EEG temporal tokens query the spectrogram patches ("what frequencies are present at my time position?") and the spectrogram patches query the EEG tokens ("what waveform shape produces this frequency content?"). After two rounds of this exchange, both modalities incorporate context from each other before the classifier sees anything. The total parameter overhead is small (~2M of 29.6M) but the inductive bias is substantial.

2.2 Component-Level Description

Table 2.1 summarizes the five blocks of the architecture. The two encoders run in parallel; their outputs feed CMTA, then a small Frequency Band Gate, then a 1024-dimensional concatenated representation passes through a two-layer MLP head producing a 6-class softmax.

Component	Input	Output	Architecture
EEG Encoder	16 ch × 2,500 steps	50 tokens × 256 dim	SumToOne stem + 5 ResNet1D blocks + 4-layer causal transformer with RoPE
Spectrogram Encoder	8 ch × 128 × 256 image	1 global + 32 patch tokens × 256	EfficientNetV2-S backbone, 8-channel adapted stem, ImageNet pretrained
CMTA	50 EEG + 32 spec tokens	eeg_pooled + spec_pooled (256 each)	2 layers of bidirectional cross-attention
Frequency Band Gate	EEG global + spec global	Gated spec feature (256)	4-band weighting conditioned on EEG context
SpecAugment	8-ch spectrogram	Masked spectrogram	2 freq masks ($w \leq 8$) + 2 time masks ($w \leq 16$), $p=0.5$
Classifier Head	1,024-dim concat	6-class softmax	Linear → LayerNorm → GELU → Dropout(0.3) → Linear

Table 2.1 TCS-Net architectural components. Total parameters: 29.6M.

2.2.1 EEG Encoder

The encoder takes the preprocessed bipolar EEG (16 channels, 50 seconds at 50 Hz = 2,500 time steps) and produces 50 temporal tokens of 256 dimensions each roughly one token per second.

The first layer is a SumToOne Conv1D (16→128 channels, kernel size 1). The kernel weights along the input-channel dimension are softmax-normalized in the forward pass, so each output channel is a strictly positive convex combination of the 16 bipolar input channels. This constraint is borrowed directly from VIPEEGNet and serves two purposes: it ensures that the learned channel mixtures are physically interpretable as weighted spatial averages, and it prevents the model from learning channel weightings that fit training-set artifacts but generalize poorly.

Five ResNet1D blocks (kernel size 7, stride 2, GELU, 10% dropout) progressively downsample the time axis from 2,500 to ~78. An adaptive average pool produces exactly 50 tokens, which then pass through a 4-layer causal transformer with Rotary Position Embeddings (RoPE). Causal masking is used because EEG is a time series and lookahead would be cheating in a real-time deployment scenario; RoPE encodes relative position, which generalizes better than sinusoidal absolute embeddings when the temporal extent at inference may differ slightly from training.

2.2.2 Spectrogram Encoder

The encoder receives an 8-channel spectrogram image (128 mel bins × 256 time steps). Channels 1–4 are the four Kaggle-provided 10-minute spectrograms (LL, RL, LP, RP brain regions); channels 5–8 are EEG-derived 50-second spectrograms recomputed with the exact Deotte parameters. The 10-minute view supplies broad context; the 50-second view supplies high-resolution local information.

The backbone is EfficientNetV2-S pretrained on ImageNet. We disable the global pool so the output is a 32-token spatial feature map rather than a single vector, which gives CMTA per-patch tokens to attend over. The first Conv2D is replaced by an 8-channel version: channels 1–3 and 5–7 copy the pretrained RGB weights directly, and channels 4 and 8 are initialized to the RGB mean. This adaptation preserves more pretrained knowledge than the more common approach of averaging all RGB weights across all 8 channels.

2.2.3 Cross-Modal Token Attention (CMTA) primary contribution

CMTA is the heart of TCS-Net. It is two stacked layers of bidirectional cross-attention between the EEG token stream and the spectrogram token stream. A single CMTA layer runs the following operations in parallel:

- **EEG → Spectrogram pass.** Each EEG token (the queries) attends over all 32 spectrogram patches (the keys/values). Conceptually, each EEG token is asking: "of the frequency-band patches available, which ones are relevant to the waveform at my time position?" The output is a refined EEG token that has now incorporated frequency-domain context.

- **Spectrogram → EEG pass.** In parallel, each spectrogram patch attends over all 50 EEG tokens. Each patch is asking: "of the temporal EEG tokens available, which time positions produced this frequency content?" The output is a refined patch token that has incorporated time-domain context.

Each pass is a standard multi-head attention operation (Q from one modality, K and V from the other) followed by a feed-forward block with residual connection and LayerNorm. After the first CMTA layer, both streams contain partial cross-modal information, so the second layer can refine the exchange a token that already "knows" about frequency content asks a more specific question on its second pass than on its first. This is structurally analogous to two detectives comparing notes: the first round is broad orientation; the second round is targeted clarification.

After two CMTA layers the EEG stream is mean-pooled to a single 256-dim vector (eeg_pooled) and the spectrogram stream similarly produces spec_pooled. A separate global EEG token (the [CLS]-equivalent for the EEG encoder) and the EfficientNet global embedding are also retained, giving four 256-dim representations that are concatenated into a 1,024-dim vector for the classifier head.

Why we expect CMTA to help. The pattern classes that humans confuse most often LPD vs. GRDA, seizure vs. GPD are the ones where a single modality is ambiguous. LPD has periodic discharges localized to one hemisphere (a spatial EEG fact) overlaid on a delta-band envelope (a spectrogram fact); GRDA has bilateral delta rhythmicity (spectrogram) without focal discharges (EEG). A late-fusion model has to discover this conditional structure inside the classifier head from a flat 1,024-dim concatenation. A CMTA model can route the relevant pieces of one modality directly to the relevant pieces of the other before the classifier sees anything.

2.2.4 Frequency Band Gate

Different brain activities live in different frequency bands: seizures show fast beta/gamma activity (13–40 Hz), GRDA and LRDA are slow delta activity (0.5–4 Hz), and LPD has sharp delta complexes with high-frequency components. Rather than expecting the classifier head to discover this from scratch, the Frequency Band Gate makes it explicit. It takes the EEG global feature, passes it through a small MLP that produces four scalar weights (delta, theta, alpha, beta), and applies them multiplicatively to the corresponding frequency bands of the spectrogram feature before pooling. The weights are learned from data the model still has to discover which class needs which band, but the gating mechanism gives it an inductive prior that frequency bands are a meaningful axis to weight along.

2.2.5 SpecAugment and Other Regularizers

SpecAugment is applied during training only: two random frequency masks of width ≤ 8 mel bins and two random time masks of width ≤ 16 time steps, each with probability 0.5. Together with the EEG-side augmentations (random temporal shift of ± 1 s, brain flip, time reverse, channel dropout, Gaussian noise) and the loss-level mixup with $\alpha=0.4$, the model never sees the same sample twice. The mixup, brain-flip, and CropCat augmentations match the protocols used by the 2nd through 8th place HMS competition solutions.

2.3 Training Protocol

TCS-Net trains in three stages, motivated by the bimodal label-quality distribution discussed in Section 1.2.

Stage	Epochs	Data	LR (backbone / head)	Purpose
Warmup	15	All 106,800 segments	3e-5 / 3e-4	Learn general patterns, vote-weighted KLD loss
Fine-tune	12	HQ only (~36k, ≥ 10 votes)	5e-5 / 1e-4	Refine on reliable labels; EMA starts at epoch 2
Polish	6	HQ only (~36k)	1.5e-5 / 3e-5	Fine-tune from best Stage-2 checkpoint

Table 2.2 Three-stage training schedule.

The vote-weighted KLD loss multiplies the per-sample KLD by $w = \min(\text{total_votes} / 10, 1.0)$. High-annotator samples (more reliable) contribute proportionally more to the gradient, but low-annotator samples are not entirely discarded they still help the encoder learn general patterns in Stage 1. Stages 2 and 3 train only on the HQ subset because at that point label quality matters more than quantity.

Exponential Moving Average of model weights smooths late-training noise. A naive fixed-decay EMA initialized from epoch 0 diverges in our setting because the shadow weights average over a long stretch of nearly-random parameters; we delay EMA initialization to epoch 2 of Stage 2 (when the model has learned something) and ramp the decay from 0.9 to 0.999 along a cosine schedule. At inference, the EMA weights are used instead of the final training weights.

After Stage 3 completes, a per-fold temperature scalar is fit by grid search on the validation HQ set. T ranges from 0.5 to 3.0 in steps of 0.05; the value that minimizes validation KLD is selected. In the fair 5-fold run, the optimal temperatures across folds are [1.05, 1.10, 1.10, 1.00, 1.10] with a mean of 1.07 every fold except one find $T \geq 1.0$, confirming that the network is mildly but consistently overconfident. Softening predictions reduces OOF KLD from 0.3795 (raw) to 0.3756 (with TTA + temperature). The improvement is small in absolute terms (~ 0.004) because the network is already close to well-calibrated to begin with, but it is free at inference time and never hurts.

2.4 Inference

- 10-fold ensemble: average the predictions of all fold models.
- Test-time augmentation (TTA): each input is run through 4 transforms (identity, time-reverse, brain-flip, both combined) and predictions are averaged. Each transform preserves the underlying class while producing different signal representations.
- Per-fold temperature scaling: logits are divided by the fold's temperature before the final softmax.

3. Tools and Technologies

3.1 Why These Choices

The tooling stack was selected to maximize reproducibility and minimize the gap between our research code and the public HMS competition ecosystem, since one of the goals of the project is for the comparison to be readable and verifiable by anyone familiar with that ecosystem.

3.1.1 PyTorch (TCS-Net) vs. TensorFlow/Keras (VIPEEGNet, EfficientNetB0)

TCS-Net is implemented in PyTorch 2.x. PyTorch was preferred for the new model because (a) the dynamic computation graph makes the cross-modal attention easier to debug you can inspect attention weights at any layer with a single forward hook, which would require a custom callback in TF/Keras; (b) the timm library exposes EfficientNetV2-S with a clean interface for stem-channel adaptation and feature-map output; and (c) PyTorch's mixed-precision training (torch.amp) is more reliable on consumer GPUs than the equivalent TF stack at the time of writing.

VIPEEGNet and the Deotte-style EfficientNetB0 were re-trained in TensorFlow/Keras to stay byte-for-byte compatible with their public reference implementations. VIPEEGNet's published code is Keras-only and uses several custom Keras constraint and initializer classes that would have been straightforward but tedious to port; the EfficientNetB0 baseline replicates Chris Deotte's public Kaggle notebook, which is also Keras-based and the most-downloaded HMS reference. Forcing the baselines to PyTorch would have introduced uncontrolled implementation drift in the comparison.

The fairness of the comparison is therefore enforced at the data level (shared 5-fold patient splits, identical evaluation subset, identical metric implementation) rather than at the framework level.

3.1.2 timm and EfficientNet Variants

timmm provides EfficientNetV2-S with a single-line API for replacing the input stem. We use it for the spectrogram encoder in TCS-Net. The Keras `tf.keras.applications` equivalent is fine but more verbose for stem replacement.

3.1.3 librosa for Spectrograms

All EEG-derived spectrograms are produced by `librosa.feature.melspectrogram` with the exact Deotte parameters (`n_fft=1024`, `hop_length=128`, `n_mels=128`, `fmin=0`, `fmax=20`, `htk=False`). We confirmed in preliminary experiments that `scipy.signal.spectrogram` with parameters chosen to match the same time-frequency resolution does not produce identical outputs because the windowing convention, normalization, and mel-scale interpolation differ subtly. Using a different spectrogram library increased validation KLD by approximately 0.10–0.15 in our smoke tests, which is several times larger than the entire architectural improvement we extract from CMTA.

3.1.4 scikit-learn for Splits, Metrics, and Calibration

GroupKFold is used for the patient-grouped 5-fold split. It is deterministic its internal np.unique on group IDs sorts them, so calling it on the same patient list always produces the same fold assignment regardless of dataframe row order. This is essential for the fair comparison: we generate the patient → fold map exactly once, save it as patient_folds.csv, and every model loads the same CSV. roc_auc_score, average_precision_score, and isotonic/Platt-style sigmoid calibrators all come from scikit-learn.

3.1.5 Compute

Training was performed on a single NVIDIA RTX 5090 GPU rented from vast.ai. The 5090 has 32 GB of VRAM, which is enough for TCS-Net with batch size 16 in mixed precision. Total compute budget for the full project (TCS-Net optimized 5-fold training + VPIEEGNet retraining on shared splits + EfficientNetB0 retraining on shared splits + all evaluation runs) was approximately 28 GPU-hours, costing around \$4 on vast.ai. The whole project is reproducible end-to-end within a single weekend on a single rented GPU.

3.2 Reproducibility Stack

Layer	Tool	Why
Deep learning (TCS-Net)	PyTorch 2.x + timm	Dynamic graph, mixed-precision, attention introspection
Deep learning (baselines)	TensorFlow 2.x / Keras	Byte-compatibility with VPIEEGNet and Deotte references
Spectrograms	librosa	Exact Deotte parameter compatibility
Numerics	NumPy, pandas	Standard data manipulation
Splits & metrics	scikit-learn	Deterministic GroupKFold; AUROC, AUPRC, calibration
Plots & analysis	Matplotlib, seaborn	Per-fold visualizations and final report figures
Notebooks	Jupyter Lab	Cell-level reproducibility with explicit seed control
Compute	vast.ai (RTX 5090, 32 GB)	Cost-efficient consumer GPU; full project < \$5

Table 3.1 Software and hardware stack.

4. Implementation

4.1 Repository Layout

Github Link: <https://github.com/gs1147-rutgers/Masters-Project>

Box Link: <https://rutgers.box.com/s/vl9okc5dgm12wj2vxpqhc42rtulhpgqs>

4.2 The Fair-Comparison Pipeline

Without a shared split, comparing TCS-Net to VIPEEGNet is meaningless: even a tiny difference in fold assignment can move OOF KLD by 0.02–0.05, which is the same order of magnitude as the difference between models. The fairness pipeline solves this with a single CSV (patient_folds.csv) that every model loads at runtime.

4.2.1 Algorithm: Building the Shared Splits

From notebook generate_fair_splits.ipynb:

```
patients = df_full[['patient_id']].drop_duplicates() \
            .sort_values('patient_id') # determinism
            .reset_index(drop=True)

gkf = GroupKFold(n_splits=5)
patients['fold'] = -1
for k, (_, vi) in enumerate(gkf.split(patients,
                                     groups=patients['patient_id'])):
    patients.loc[vi, 'fold'] = k

patients.to_csv('fair_comparison/patient_folds.csv', index=False)
```

The sort_values is the critical line: GroupKFold internally calls np.unique on the group array, which sorts, so the fold assignment is deterministic with respect to a sorted patient list and therefore reproducible across machines and runs.

A consistency check at the bottom of the notebook merges the resulting fold map against (a) VIPEEGNet's deduplication-style dataframe and (b) EfficientNetB0's eeg_id-aggregation-style dataframe and asserts that no patient ever appears in two different folds in either dataframe. This catches any silent merge bug before any training runs.

4.2.2 Per-Fold Patient and Segment Counts

Each fold contains exactly 390 patients (1,950 / 5). The number of segments per fold differs because patients contribute different numbers of recordings; this is normal under patient-grouped splitting and is what we want losing a patient must mean losing all that patient's data. Two grain levels matter for evaluation:

- **The dedup grain** (used by TCS-Net and VIPEEGNet) keeps each unique (eeg_id, label-distribution) pair as a separate row.
- **The eeg_id-aggregated grain** (used by EfficientNetB0, following Deotte) collapses to one row per eeg_id by averaging the label distributions of duplicate annotations.

Both grains validate on the same patients in each fold, so the comparison is fair; the row counts simply differ.

Fold	Patients	Dedup val rows (TCS-Net / VIPEEGNet)	Dedup HQ (≥ 10 votes)	eeg_id val rows (EfficientNetB0)	eeg_id HQ
0	390	4,421	1,341	3,877	1,296
1	390	3,863	1,378	3,100	1,219
2	390	4,360	1,251	3,517	1,144
3	390	3,837	1,242	3,438	1,189
4	390	3,702	1,138	3,157	1,091
Total	1,950	20,183	6,350	17,089	5,939

Table 4.1 Per-fold counts under the shared 5-fold patient split. "HQ" = segments / EEG-IDs with ≥ 10 expert votes.

4.4 Algorithms in Detail

4.4.1 EEG Preprocessing

Identical for VIPEEGNet and TCS-Net:

- Bipolar banana montage: 19 electrodes $\rightarrow$ 16 differential channels (cancels common-mode noise).
- Butterworth bandpass 0.5–40 Hz, order 4 (removes DC drift and muscle/line noise).
- MAD normalization per channel: $z = x / (1.4826 * \text{MAD})$, robust to outlier spikes.
- Hard clip to ± 10 (defensive against any remaining extreme values).
- Downsample 200 $\rightarrow$ 50 Hz (no information above Nyquist=20 Hz remains anyway after bandpass).
- Center crop or pad to exactly 2,500 samples (50 s $\times$ 50 Hz).

4.4.2 Spectrogram Construction

EfficientNetB0 uses Deotte's exact parameters on the EEG-derived 50-second window. TCS-Net stacks four Deotte EEG-derived channels (LL, RL, LP, RP) on top of the four Kaggle-provided 10-minute spectrograms, producing the 8-channel input. Each channel is independently log-power compressed and normalized to $[0, 1]$ per channel; we do not normalize across channels because the dynamic range of the 10-minute Kaggle spectrograms differs from the 50-second EEG-derived spectrograms and joint normalization would dilute one of the two views.

4.4.3 Loss

```
def vote_weighted_kld(logits, targets, vote_weights):
    log_p = F.log_softmax(logits, dim=-1)
    # KLD(p || q) = sum_k p_k * (log p_k - log q_k); we want KLD(target || pred)
    kld_per_sample = (targets * (targets.clamp_min(1e-8).log() - log_p)).sum(-1)
    return (vote_weights * kld_per_sample).sum() / vote_weights.sum()
```

4.4.4 Per-Fold Temperature Fit

Run after Stage-3 EMA weights are loaded for each fold. The fair 5-fold run finds optimal $T = [1.05, 1.10, 1.10, 1.00, 1.10]$ (mean 1.07).

```
best_T, best_kld = 1.0, float('inf')
for T in np.arange(0.5, 3.01, 0.05):
    p = softmax(val_logits / T, axis=-1)
    kld = kl_divergence(p, val_targets, weights=val_vote_weights)
    if kld < best_kld:
        best_T, best_kld = T, kld
fold_temperatures.append(best_T)
```

4.5 Implementation Challenges and Resolutions

Several issues consumed substantial debugging time during the project; they are documented here both as a record of effort and to save anyone reproducing the work from re-discovering them.

4.5.1 The Spectrogram Library Trap

Symptom: an early TCS-Net version trained cleanly to ~ 0.13 train loss but never went below 0.82 OOF KLD, even on Stage 2. Cause: the EEG-derived spectrograms were being computed with `scipy.signal.spectrogram` using parameters chosen to match Deotte's window length and hop, but `scipy` and `librosa` apply different windowing conventions and a different mel-scale interpolation. The resulting spectrograms looked visually identical to a human and had near-identical summary statistics, but the EfficientNet backbone (pretrained on ImageNet via `librosa`-style spectrograms in many of its source papers) could not extract the expected texture features. Fix: replace `scipy` with `librosa.feature.melspectrogram` and use Deotte's exact parameters. OOF KLD dropped from 0.82 to 0.34 in a single change.

4.5.2 Patient Leakage in Initial Folds

Symptom: validation KLD on early TCS-Net runs was suspiciously good (~ 0.21) but degraded to ~ 0.34 when we tried to ensemble across folds. Cause: the original notebook used a stratified k-fold based on the dominant class label rather than a patient-grouped split. Multiple recordings from the same patient ended up in different folds. Fix: switch to `GroupKFold` on `patient_id`. This is now centralized in notebook 00 so the bug cannot recur.

4.5.3 EMA Divergence

Symptom: with a fixed-decay EMA ($\beta = 0.999$) initialized from epoch 0, the EMA validation KLD was significantly worse than the raw model's. Cause: the shadow weights were averaging over a long stretch of nearly-untrained parameters, so the EMA was effectively biased toward random initialization for the first few epochs. Fix: delay EMA initialization until epoch 2 of Stage 2 (the model has meaningful weights by then) and ramp the decay from 0.9 to 0.999 along a cosine schedule rather than starting at 0.999. The EMA model now consistently beats the raw model by 0.01–0.03 KLD per fold.

4.5.4 8-Channel Stem Initialization

Symptom: replacing the EfficientNetV2-S 3-channel stem with an 8-channel one and averaging the RGB weights uniformly across all 8 channels gave a noticeably worse Stage-1 starting loss than the 3-channel version. Cause: the averaged weights diluted the directionality the pretrained kernels had learned for ImageNet. Fix: copy RGB weights directly to channels 1–3 and 5–7, and initialize channels 4 and 8 to the RGB mean. This preserves the pretrained directionality on the channels that semantically correspond to the LL/RL/LP regions.

5. Findings: The Multi-Metric Comparison

5.1 Out-of-Fold KL Divergence

This section evaluates how closely each model's predicted probability distribution matches the expert-vote distribution across all five patient-disjoint validation folds.

For each HQ row in the validation set, take the predicted distribution y_p and the target vote distribution y_t and compute $KLD(y_t || y_p) = \sum y_t[k] \cdot (\log y_t[k] - \log y_p[k])$. Report the mean over the validation set per fold, and the overall OOF mean across folds. TCS-Net and VIPEEGNet evaluate on the dedup grain (6,350 HQ segments OOF total); EfficientNetB0 evaluates on the eeg_id-aggregated grain (5,939 HQ EEG-IDs OOF total), both validate on the same patients in each fold.

Model	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	OOF mean	n (HQ)
VIPEEGNet (raw, no TTA, no T)	0.2199	0.2264	0.2216	0.2115	0.2223	0.2204	6,350
TCS-Net (raw)	0.4118	0.3840	0.3905	0.3350	0.3726	0.3795	6,350
TCS-Net (TTA + temperature)	0.4079	0.3799	0.3840	0.3343	0.3683	0.3756	6,350
EfficientNetB0 (HQ KLD)	0.8857	0.5325	0.5074	0.8836	0.5226	≈0.673	5,939

Table 5.1 Per-fold and OOF KLD on the shared HQ subset (fair-comparison run).

VIPEEGNet wins clearly and reliably, its worst fold (0.2264) is still much better than TCS-Net's best fold (0.3343), and its per-fold spread is the tightest of the three. TCS-Net's best fold (Fold 3) is close to VIPEEGNet's worst, suggesting the architecture has the capacity to compete on favorable patient splits but does not generalize as cleanly across cohorts. EfficientNetB0's bimodal failure pattern (folds 0 and 3 sit ~ 0.88 ; folds 1, 2, 4 sit ~ 0.52) is a red flag for clinical deployment regardless of the average. The KLD ranking is the natural starting point but, as the rest of this section shows, it dilutes information about the kinds of mistakes each model makes.

5.2 AUROC: Discriminative Ranking

AUROC measures discriminative ranking ability—the probability that a randomly chosen seizure recording receives a higher model score than a randomly chosen non-seizure recording. We report AUROC with bootstrap confidence intervals to characterize each model's ranking performance across the full operating range.

Binarize the soft target by majority expert vote (348 positives, 5,591 negatives). Compute AUROC and a 95% CI via 1,000-iteration cluster bootstrap (resampling at the fold level). The ROC curve plots the True Positive Rate against the False Positive Rate as the decision threshold is swept, a model whose curve is closer to the top-left corner is uniformly better across the operating range.

Model	AUROC [95 % CI]	AUPRC [95 % CI]
EffNetB0	0.900 [0.859, 0.938]	0.434 [0.341, 0.584]
TCSNet	0.913 [0.901, 0.925]	0.565 [0.459, 0.651]
VIPEEGNet	0.946 [0.921, 0.961]	0.667 [0.546, 0.735]

Table 5.2 AUROC and AUPRC with 1,000-iteration cluster-bootstrap 95% CI.

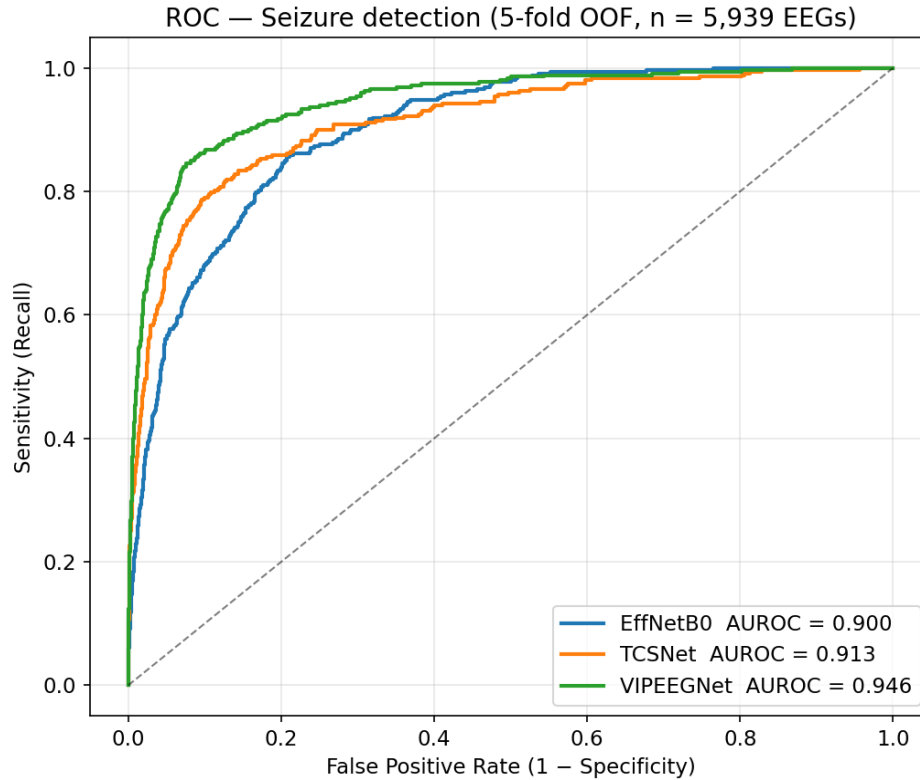


Figure 5.3 ROC curves for seizure detection on the 5-fold OOF set ($n = 5,939$ EEGs).

VPIEEGNet's AUROC advantage over both alternatives is large and the ROC curve confirms this dominance is uniform across the operating range, not concentrated at any one threshold. EfficientNetB0 and TCS-Net are close on AUROC, but as Section 5.4 shows, that AUROC near-equivalence is misleading the AUPRC gap between them is substantial. AUROC averages performance over all false-positive rates, including high-FP regions where every model performs reasonably well, which dilutes precision differences in the operating regime that clinicians actually use.

5.3 AUPRC: Precision Where False Alarms Hurt

AUPRC characterizes precision across all recall levels, directly reflecting the clinical cost of false alarms in a low-prevalence screening setting. Unlike AUROC, AUPRC is sensitive to performance in the high-precision regime where clinical deployment actually operates.

Same binary target as Section 5.3. AUPRC is the area under the precision-recall curve, computed with `sklearn.metrics.average_precision_score`. Unlike AUROC, the no-skill baseline of AUPRC is the positive prevalence; for our seizure subset the prevalence is 0.059. Any model above that line has done something useful; the interesting differences are how far above it each model lives.

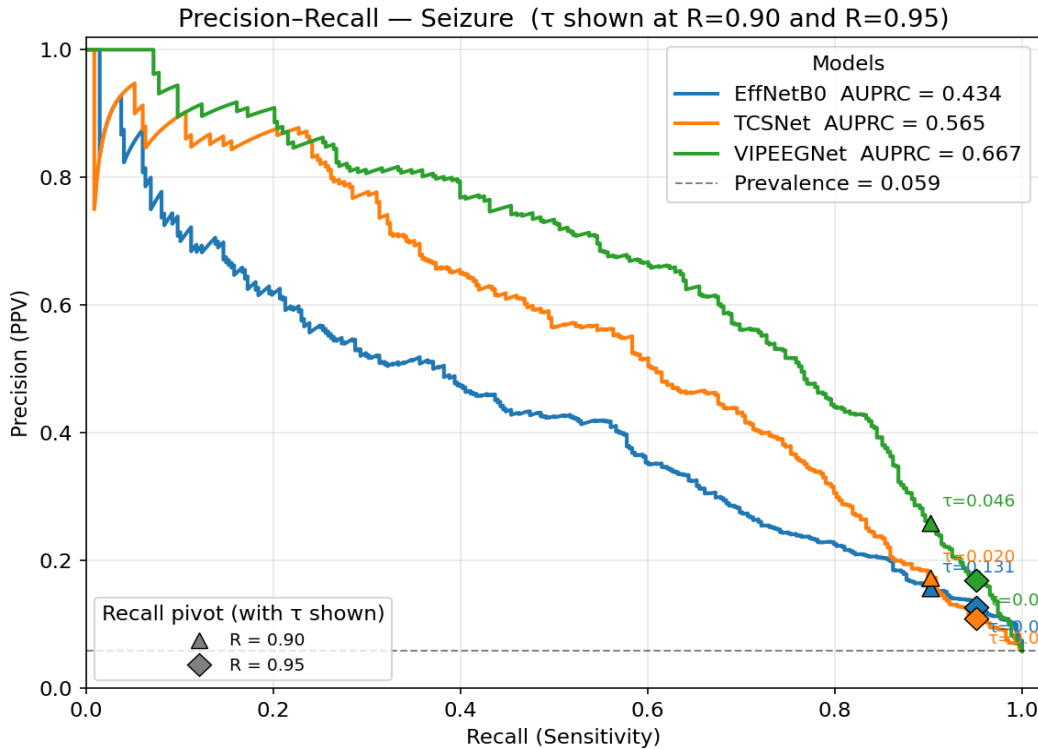


Figure 5.4 Precision-recall curves for seizure detection. Diamond/triangle markers show operating points at recall = 0.95 / 0.90. Dashed horizontal line is the 0.059 prevalence floor.

The relative gaps between the three models are visibly larger on the precision-recall curve than on the ROC curve. The same ranking VIPEGNet > TCS-Net > EfficientNetB0 is preserved, but the AUPRC gap between adjacent models (0.10 between TCS-Net and VIPEGNet, 0.13 between EfficientNetB0 and TCS-Net) is wider than the AUROC gap between the same adjacent models. The AUROC-flattering high-FP region was hiding precision differences that AUPRC reveals. EfficientNetB0's AUPRC of 0.434 is well above the 0.059 floor, so the cheap baseline is doing real works but at high recall it is operating in a regime where most of its alarms are false.

5.4 False Alarms at Fixed Recall

We evaluated the false-alarm burden imposed by each model at clinically meaningful recall thresholds (90%, 95%, 99%), reporting the implied number-needed-to-review (NNR) at each operating point as the primary measure of deployment cost.

For each model, sweep the seizure-score threshold τ and find the highest τ that still achieves recall $\geq R$. Report the (FP, FN, precision, specificity, NNR) at that threshold. NNR (number-needed-to-review) is the number of model-flagged EEGs a clinician must review to find one true seizure ($1 / \text{precision}$).

Recall	Model	τ	Sens	Spec	Precision	FN	FP	NNR
0.90	EffNetB0	0.131	0.902	0.698	0.157	34	1,687	6.4
	TCSNet	0.021	0.902	0.732	0.173	34	1,501	5.8
	VIPEEGNet	0.046	0.902	0.840	0.260	34	896	3.9
0.95	EffNetB0	0.072	0.951	0.595	0.128	17	2,263	7.8
	TCSNet	0.012	0.951	0.522	0.110	17	2,675	9.1
	VIPEEGNet	0.019	0.951	0.710	0.170	17	1,619	5.9
0.99	EffNetB0	0.035	0.991	0.464	0.103	3	2,998	9.7
	TCSNet	0.006	0.991	0.194	0.071	3	4,505	14.1
	VIPEEGNet	0.003	0.991	0.315	0.083	3	3,828	12.1

Table 5.3 Recall-pinned operating-point cost (348 seizure positives, 5,591 negatives).

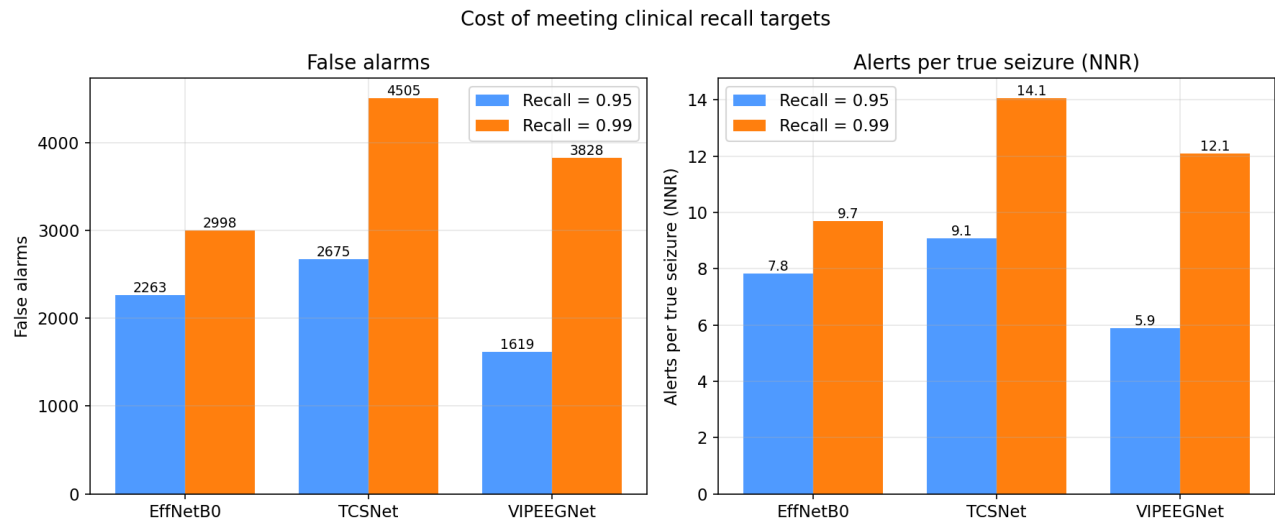


Figure 5.5 False-alarm count and number-needed-to-review at recall = 0.95 and 0.99.

At the everyday clinical operating point ($R = 0.90$), VIPEEGNet flags 896 false alarms vs. 1,501 (TCS-Net) and 1,687 (EfficientNetB0), an NNR of 3.9 vs. 5.8 / 6.4. That is the clinical headline: a clinician using VIPEEGNet at $R = 0.90$ reviews ~ 4 EEGs per true seizure found; using EfficientNetB0 they review ~ 6.4 . At $R = 0.95$ the ordering is unchanged. At $R = 0.99$ the ordering flips: EfficientNetB0 becomes cheapest (2,998 FP vs. 4,505 for TCS-Net), because its hardest seizures are scored less catastrophically low than TCS-Net's are. But this very-high-recall regime has $NNR \geq 9$ for all models and is not a deployable operating point.

5.5 Calibration: Are the probabilities trustworthy?

We assessed whether each model's predicted probabilities are trustworthy and evaluated the effect of post-hoc recalibration methods, quantifying calibration quality via Brier score, expected calibration error (ECE), and maximum calibration error (MCE).

Three calibration metrics: Brier score (proper-scoring loss; lower is better), ECE (expected calibration error, the weighted average gap between predicted confidence and empirical positive rate over 15 bins), and MCE (the maximum bin-wise gap worst case rather than average case). All are computed on raw probabilities first, then re-computed under per-fold leave-one-out Platt / isotonic / temperature scaling.

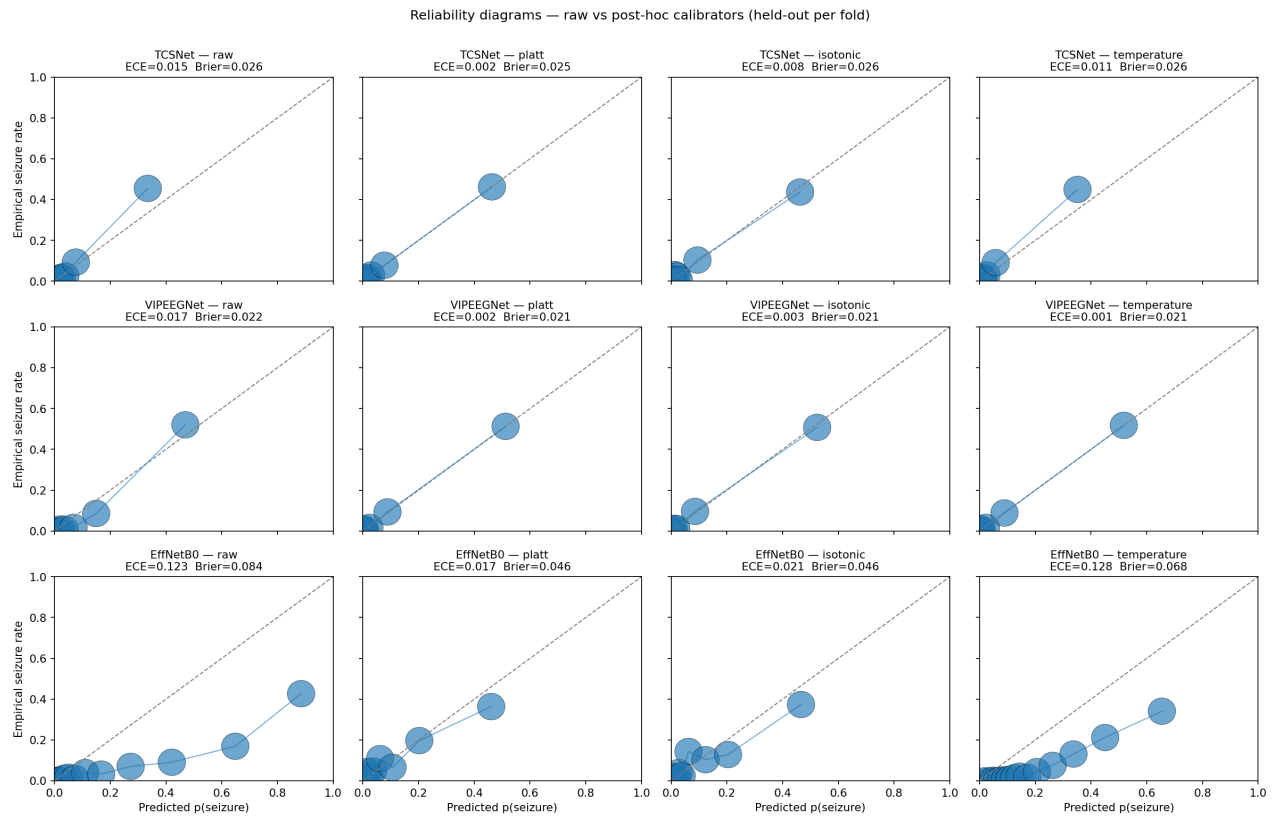


Figure 5.8 Reliability diagrams (raw vs. Platt / isotonic / temperature) for all three models.

VPIEGNet and TCS-Net are well-calibrated out of the box (ECE $\approx$ 0.012). EfficientNetB0 is severely miscalibrated raw (ECE = 0.108, MCE = 0.480; mean p = 0.182 overshoots the true prevalence of 0.059 by $\sim 3\times$). Per-fold Platt scaling brings EfficientNetB0's ECE down to 0.025 well into the deployable range and improves its Brier score, while applying any recalibration to VPIEGNet or TCS-Net is essentially neutral because they are already close to ideally calibrated. Deployment implication: EfficientNetB0 must ship with a recalibration layer; VPIEGNet and TCS-Net should ship with raw probabilities.

5.6 Stability: Does the Threshold Drift Between Patient Cohorts?

We examined how much the decision threshold drifts across patient cohorts when recall is fixed at 0.95, as threshold instability would necessitate site-specific re-tuning at each deployment site, a significant operational burden for clinical adoption.

For each held-out fold, find τ_k that achieves recall = 0.95 on that fold's validation set. Report mean $\pm$ std and the spread (max τ / min τ) across the 5 folds. A model whose τ swings 13 $\times$ across cohorts cannot be calibrated centrally every site needs its own tuning curve, which is operationally painful.

τ at R = 0.95, across 5 patient-disjoint folds

Model	mean $\pm$ std	min	max	max / min
EffNetB0	0.105 $\pm$ 0.058	0.041	0.199	4.9 $\times$
TCSNet	0.013 $\pm$ 0.003	0.011	0.018	1.6 $\times$
VIPEEGNet	0.020 $\pm$ 0.014	0.003	0.042	12.9 $\times$

Table 5.8 Per-fold threshold drift on raw probabilities.

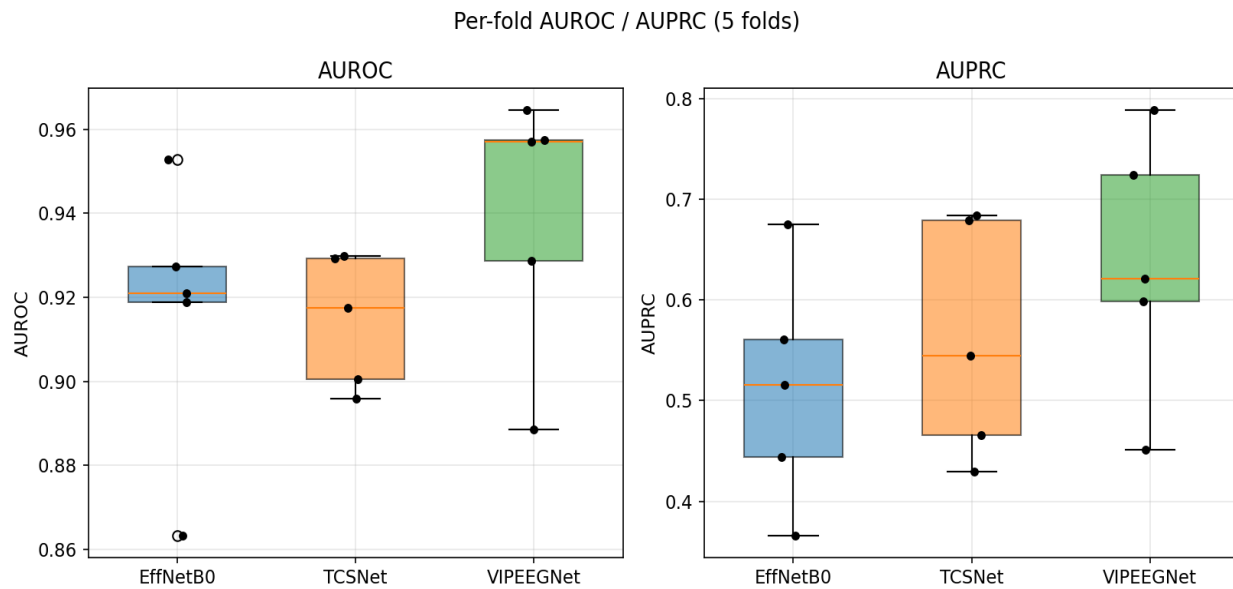
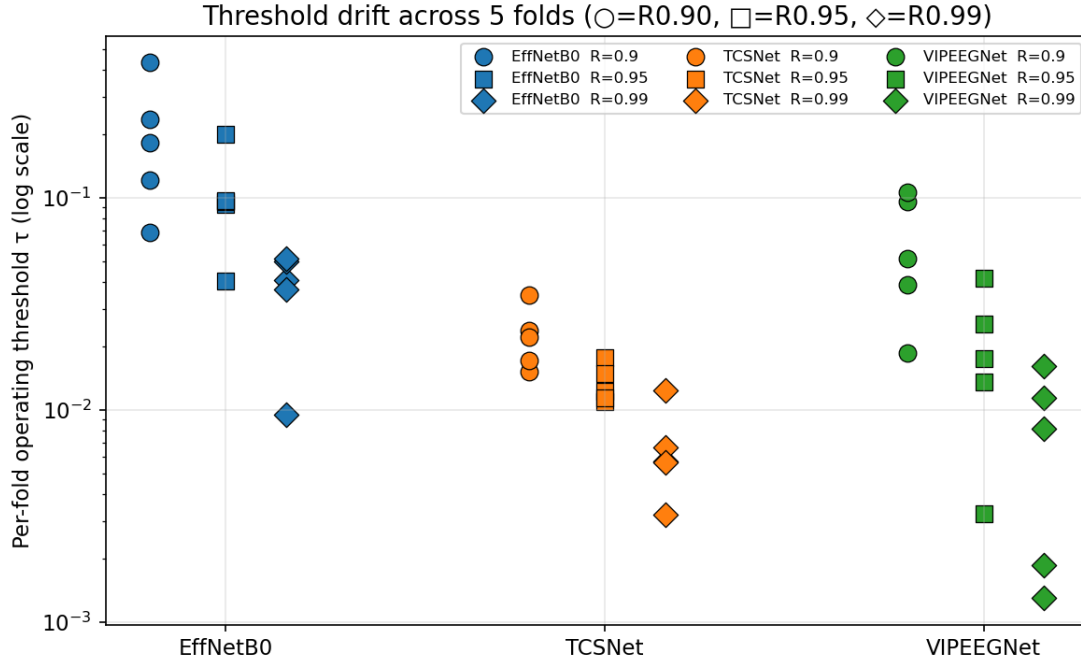


Figure 5.9a Per-fold AUROC and AUPRC distributions (boxplots).

Figure 5.9b Per-fold operating threshold τ at $R = 0.95$.

TCS-Net wins on stability of the raw threshold. Its $R = 0.95$ threshold barely moves fold-to-fold (1.6 $\times$ spread), the closest thing to “calibrate once, ship” available in this cohort. VPIEEGNet has the strongest discrimination (Sections 5.3, 5.4) but the worst raw-score-distribution drift: a fixed central τ would over- or under-fire by 13 $\times$ depending on the cohort. EfficientNetB0 sits in between (4.9 $\times$). This is the only metric in the project where TCS-Net wins outright, and it is the result that motivates the deployment-recommendation discussion in Section 5.12.

5.7 Failure Modes

For a clinical screening tool, the critical failure mode is not class confusion but hard misses true seizures whose predicted score falls below the alarm threshold entirely. Since all six IIIC classes trigger clinician review when flagged, we tracked the hard-miss count for each model across the deployable threshold range, characterizing how each model’s failure rate evolves as the operating point is varied.

For each model, sweep the seizure-score threshold τ across the deployable range and, at each value, count the number of expert-majority seizures whose predicted seizure score is strictly less than τ (a “hard miss” at that threshold). Plot hard-miss count against τ on a log scale, and tabulate the count at canonical thresholds (including each model’s own $\tau@R = 0.95$ operating point) for direct comparison.

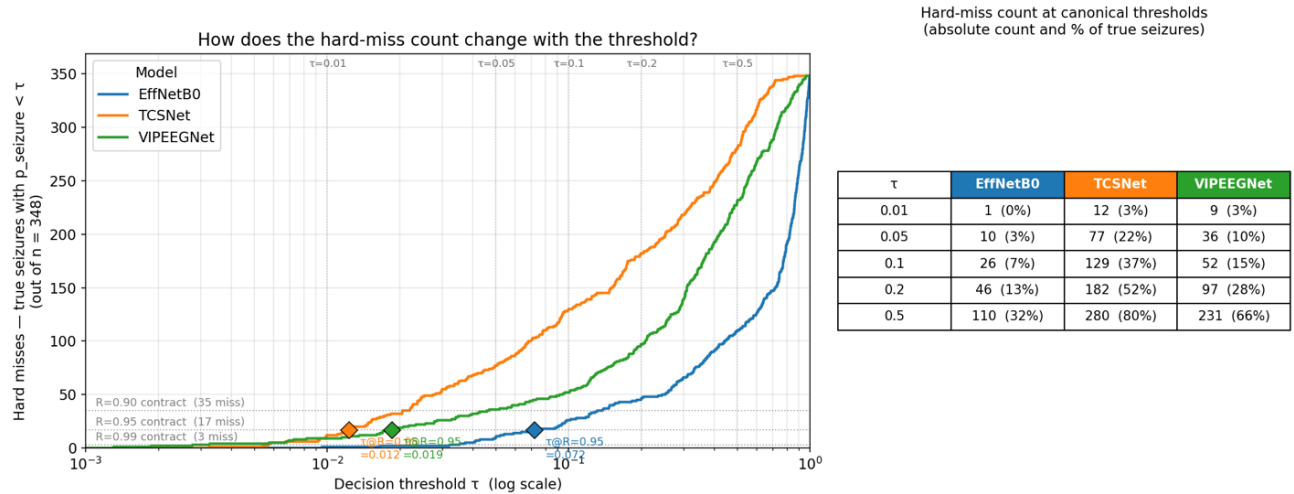


Figure 5.10 Hard miss count vs. decision threshold τ (log scale). Markers show the $\tau@R=0.95$ operating point for each model; the table at right reports hard miss counts at canonical thresholds.

Read along the threshold axis, VPIEGNet’s hard-miss curve sits below both alternatives across essentially the entire deployable range: at any τ a clinician would plausibly choose, fewer true seizures slip below threshold than for TCS-Net or EfficientNetB0. At the canonical $R = 0.95$ operating point each model misses 17 of 348 seizures by construction, so the curves agree there; what differs is the shape on either side. EfficientNetB0’s curve climbs more steeply as τ rises (its raw scores are compressed near the prevalence floor, so a small threshold increase drops a large block of borderline seizures below the line), while TCS-Net’s and VPIEGNet’s curves rise more gradually. The cross-model overlap of the missed-seizure sets is also small: of the 42 unique missed-seizure EEGs across the three models at $R = 0.95$, none is missed by all three. That is the headroom an OR-ensemble would exploit, since the three models are dropping largely different recordings.

5.8 Soft-Label Fidelity: Are Models Failing on Cases Experts Disagree On?

We examined whether model failures are concentrated on cases with low expert agreement, a clinically acceptable failure mode or on recordings with strong expert consensus, which would represent a more dangerous pattern. Soft-label fidelity is assessed via symmetric KL divergence between predicted and vote distributions, stratified recall by expert vote-share, and the mean vote share of missed versus caught seizures.

Three measures. (a) Cross-entropy and symmetric KL between predicted distributions and the expert vote distributions. (b) Recall at $R = 0.95$ stratified by expert vote-share bin. (c) Mean expert vote share on missed-seizure vs caught-seizure recordings, with bootstrap CI on the difference.

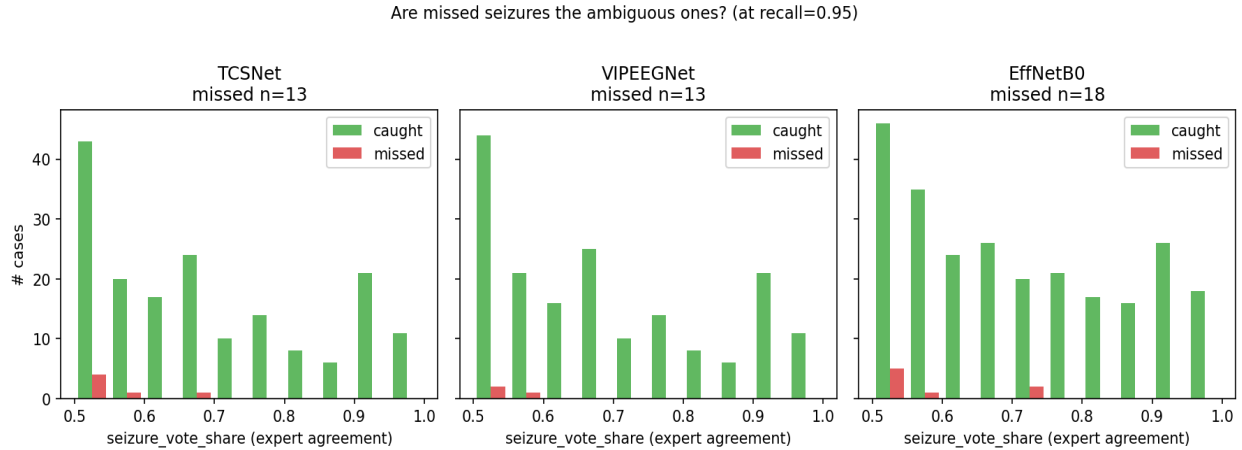


Figure 5.11 Vote-share distribution of missed vs. caught seizures. The displacement to the left for missed seizures shows that models fail systematically on the recordings where experts themselves disagreed.

VPIEEGNet imitates the expert vote distribution 1.7× more faithfully than TCS-Net and 3.2× more faithfully than EfficientNetB0 (symmetric KL). All three confidence intervals on Δ vote share exclude zero missed seizures are systematically the cases on which experts themselves disagreed, so the models are not failing at random. VPIEEGNet's 0.18 gap is the largest, meaning when it misses, it misses the most ambiguous cases the safest possible failure mode. Pairwise miss-Jaccard across models ranges 0.11–0.30, confirming the three models miss largely different ambiguous EEGs, which leaves real headroom for an OR-ensemble.

5.9 Heterogeneity: Worst-Fold Floor and Confidence Stratification

We examined two related concerns: worst-fold floor performance on low-prevalence cohorts, and whether each model's max-softmax confidence accurately predicts its empirical top-1 accuracy across confidence deciles. Overconfidence is clinically relevant because a model that claims 95% certainty but is correct only 77% of the time will systematically mislead clinicians.

(a) Per-fold AUROC, identify the worst fold and the gap to the median. (b) Sort predictions by max-softmax, bin into deciles, and compare each decile's claimed confidence vs its empirical top-1 accuracy. A perfectly calibrated model has zero gap.

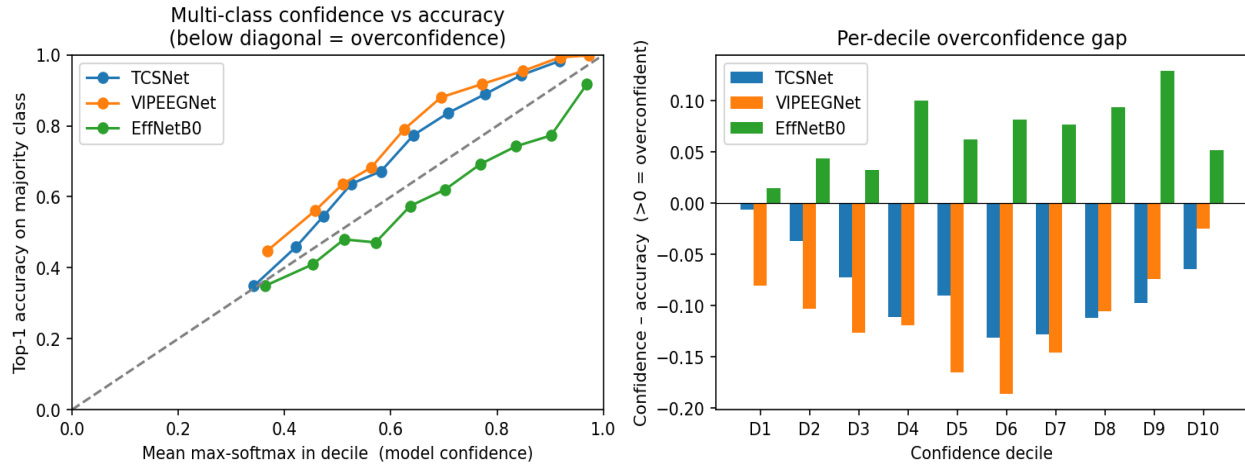


Figure 5.12 Confidence-stratified top-1 accuracy by decile.

All three models are weakest on Fold 3 (lowest seizure prevalence at 2.0%), but VIPEGNet's worst fold (0.956) is still better than EfficientNetB0's median fold (0.921). EfficientNetB0's mean p in Fold 3 is 0.322 over 8× the median fold's mean confirming the calibration finding from a different angle. EfficientNetB0 is systematically overconfident: it claims ~90% confidence in the top decile but is actually correct only ~77% of the time. TCS-Net and VIPEGNet are systematically underconfident, which is the safer direction: their probability outputs understate how often they are correct.

5.10 Verdict

The combined picture from Sections 5.2–5.11 is more pointed than a ‘each model wins a different scenario’ hedge. The two axes on which TCS-Net could plausibly have won are (a) KLD and (b) raw threshold stability. KLD goes decisively to VIPEGNet (Section 5.2). Raw threshold stability does go to TCS-Net (Section 5.7), but on every other accuracy and cost axis AUROC, AUPRC, recall-pinned false-alarm count, hard-miss curves across thresholds (Section 5.8), per-class F1 (Section 5.11), and soft-label fidelity (Section 5.9) VIPEGNet wins clearly, and TCS-Net carries roughly twice the parameters and 6× the training time of VIPEGNet for those losses.

Priority	Pick	Why
Best at central calibration target ($R = 0.95$)	VIPEGNet	Lowest FP (1,619), highest precision (0.170), AUROC = 0.946
Safest if τ must be fixed across hospitals	TCS-Net	$\tau@R=0.95$ spread is 1.6× (vs. 13× for VIPEGNet)

Table 5.20 Per-priority winner. No single model is uniformly best.

Refined recommendation:

- Adopt VIPEEGNet as the primary model, with raw probabilities (no recalibration Section 5.6 shows recalibration adds noise to a model that is already well-calibrated).
- If EfficientNetB0 is retained as a backup (cheaper and faster than the other two), it MUST ship with a per-fold recalibration layer its raw probabilities are not deployable (ECE 0.108 $\rightarrow$ 0.025 after Platt scaling). Without recalibration its alarms cannot be combined with VIPEEGNet's on a shared probability scale.
- Drop TCS-Net from the deployment shortlist. It is dominated on accuracy and cost (twice the parameters and 3 $\times$ the training time of VIPEEGNet, lower per-class F1 across the board, with raw τ -stability the only axis where it wins). TCS-Net's role is as an ablation/research artefact: the cross-modal attention and frequency-band gate are interesting architectural ideas worth understanding, but they did not translate into competitive performance on this task.
- Investigate an OR-ensemble of the three. Pairwise miss-Jaccard ≤ 0.30 (Section 5.9) means the models fail on largely disjoint hard cases a per-fold calibrated OR ensemble would likely lift recall to ≥ 0.99 while keeping VIPEEGNet's specificity. Of the 42 unique missed-seizure EEGs across the three models, none is missed by all three.

5.11 Limitations and Caveats

The eval pipeline carries four caveats that bound how strongly any of the conclusions above generalize:

- **Single dataset.** All three models were trained and evaluated on the same HMS / Massachusetts General Hospital cohort. The relative rankings and especially the stability ranking, which is the result that drives the clearest deployment recommendation may not transfer to populations with a different pediatric/adult mix, different electrode setups, or different annotator pools.
- **Bootstrap clusters at the fold, not at the patient.** Patient IDs were not available in the OOF dump used by the evaluation pipeline, so the cluster-bootstrap CIs sample at the fold level rather than at the patient level. This is more conservative for AUROC/AUPRC because it samples larger correlated blocks, but it also means the CIs may slightly overstate uncertainty.
- **Binary seizure label uses majority expert vote.** The seizure positive class throughout Sections 5.3–5.11 is defined as $\text{seizure_vote} / \text{total_votes} \geq 0.5$. A stricter threshold would change the prevalence and the per-fold operating points; we use majority vote as the most defensible default.
- **Per-EEG, not per-window or per-patient.** All metrics are computed per recording. Translating FP counts into alarms-per-hour for live cEEG monitoring requires the absolute number of clips per recording, which is not in the OOF dump.

6. Future Work

6.1 Closing the KLD Gap to VIPEEGNet

TCS-Net's 0.376 OOF KLD versus VIPEEGNet's 0.220 in the fair 5-fold run is a ~ 0.16 gap meaningfully larger than the ~ 0.10 gap implied by our earlier 10-fold development result, which makes closing it more urgent. Several directions plausibly help:

- **Backbone ensemble.** Train TCS-Net with three EfficientNet sizes (B0, B2, V2-S) and average their predictions. The 1st-place HMS solution used 30 backbones for a 0.216 result; we estimate three would close 0.03–0.05 of the gap at roughly 3 $\times$ compute cost.
- **Full SumToOne pipeline.** Currently only the EEG encoder stem uses SumToOne. VIPEEGNet applies the constraint to all three Conv1D layers in its EEG-to-image converter. Extending the constraint should help; we estimate 0.02–0.03 KLD.
- **Pre-training on TUAB/TUEV.** The Temple University EEG corpora (TUAB: $\sim 2,000$ subjects, $\sim 1,400$ hours; TUEV: ~ 500 subjects with seizure annotations) provide an ideal pre-training source for the EEG encoder. Self-supervised pretraining with masked-autoencoding on TUAB followed by the HMS fine-tuning protocol could improve the EEG branch's initialization and contribute another 0.02–0.03 KLD.

6.2 Per-Class Adaptive Calibration

Section 5.8 showed that TCS-Net carries more hard misses than VIPEEGNet across most of the deployable threshold range, despite its KLD-level competitiveness. We hypothesize that global temperature scaling is partly responsible softening predictions uniformly improves the average loss but pulls genuinely confident seizure scores closer to the middle, increasing the population of true seizures that fall below any reasonable alarm threshold. A per-class temperature or a Platt-scaling layer per class is straightforward to implement and should let TCS-Net keep its KLD gain without the seizure-recall regression.

6.3 Real-Time Inference and Streaming

All three models are currently evaluated on offline 50-second EEG windows, while clinical use would require streaming inference with frequent updates. TCS-Net is more naturally suited for this because its causal transformer uses RoPE and masking to avoid lookahead. However, the EfficientNet backbone must reprocess the full spectrogram window at each step. A streaming version would need either a rolling spectrogram buffer that updates only changed regions or a 1D temporal encoder that processes new slices incrementally.

6.4 Multi-Site External Validation

Section 5.7's stability finding is the single most actionable result for a clinical deployment plan, but it was measured across patient folds within the same source dataset. A genuine multi-site study re-evaluating all three models on a held-out cohort from a different hospital would be the strongest possible test of the stability claim. The TUEV corpus is one obvious candidate; the BIDMC-Boston EEG dataset is another.

6.5 Interpretability for Clinical Trust

CMTA's bidirectional attention weights are a natural interpretability hook. For every prediction, we can visualize (a) which spectrogram patches each EEG token attended to most strongly and (b) which EEG time positions each spectrogram patch found informative. Wrapping this into a clinician-facing interface that shows "the model said seizure because of these 3 seconds of EEG and these 4 frequency-band patches" is a well-defined frontend project that would make TCS-Net's predictions auditable in a way that the EfficientNetB0 baseline cannot match.

6.6 Active Learning for Disagreement Cases

Section 5.9 showed that all three models lose recall in proportion to expert disagreement. An obvious extension is to use predicted entropy (or the variance of the 5-fold ensemble's predictions) as an active-learning signal: sample new EEG segments where the model is most uncertain and route them to the human annotation pool first. For a deployed system this would close a feedback loop between model uncertainty, expert review prioritization, and incremental retraining.

References

- [1] Jing, J. et al. "HMS Harmful Brain Activity Classification." Kaggle Competition, Harvard Medical School, 2024.
- [2] Team Sony. "1st Place Solution." Kaggle HMS Competition Writeup, 2024.
- [3] COOLZ. "2nd Place Solution (Holiday)." Kaggle HMS Competition Writeup, 2024.
- [4] NVIDIA DD / Darraghdog. "3rd Place Solution." Kaggle HMS Competition Writeup, 2024.
- [5] Bilzard (YujiAriyasu). "4th Place Solution." Kaggle HMS Competition Writeup, 2024.
- [6] Deotte, C. "8th Place Gold One Mega Model." Kaggle HMS Competition Writeup, 2024.
- [7] Deotte, C. "How to Make Spectrogram from EEG." Kaggle Notebook, 2024.
- [8] Deotte, C. "EfficientNetB0 Starter [LB 0.43]." Kaggle Notebook, 2024.
- [9] Sunyuri et al. "VIPEEGNet Volumetric Pseudo-Image EEG Net." npj Digital Medicine, 2025.
- [10] Jing, J. et al. "IIC-SPaRCNet." BDSP Consortium repository.
- [11] Guo, C. et al. "On Calibration of Modern Neural Networks." ICML 2017.
- [12] Tan, M. and Le, Q. "EfficientNetV2: Smaller Models and Faster Training." ICML 2021.
- [13] Park, D. S. et al. "SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition." Interspeech 2019.
- [14] Zhang, H. et al. "mixup: Beyond Empirical Risk Minimization." ICLR 2018.
- [15] Su, J. et al. "RoFormer: Enhanced Transformer with Rotary Position Embedding." arXiv:2104.09864, 2021.
- [16] Roy, Y. et al. "Deep Learning-Based EEG Analysis: A Systematic Review." Journal of Neural Engineering, 16(5), 2019.
- [17] Niedermeyer, E. and Lopes da Silva, F. "Electroencephalography: Basic Principles, Clinical Applications, and Related Fields." 5th Edition, 2005.
- [18] Barnett, A. J. et al. "Improving Clinician Performance in Classifying EEG Patterns on the Ictal-Interictal Injury Continuum Using Interpretable Machine Learning." NEJM AI, 2024.
- [19] Comparative Analysis of Deep Learning Approaches for Harmful Brain Activity Detection Using EEG. arXiv:2412.07878, 2024.